

NYCU-EE IC LAB – Spring 2025

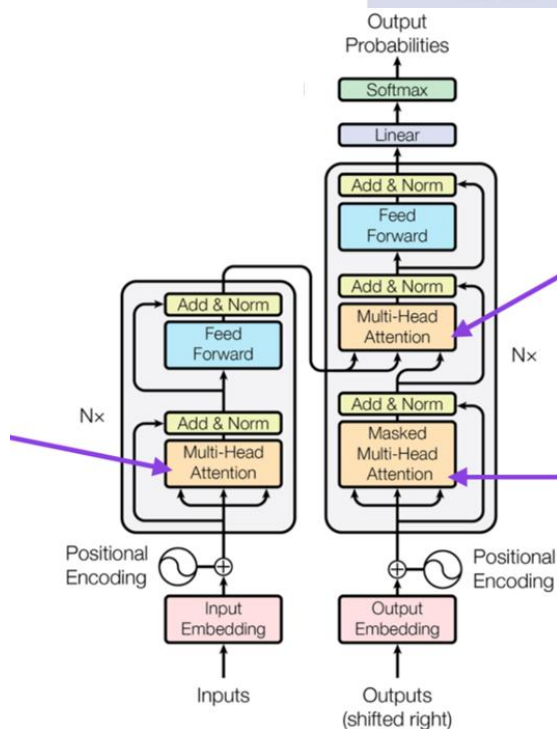
Lab04 Exercise

Design: Two Head Attention

Data Preparation

1. Extract file from TA's directory:
`% tar xvf ~iclabTA01/Lab04_S_2025.tar`
2. The extracted LAB directory contains:
 - a. **00_TESTBED**
 - b. **01_RTL**
 - c. **02_SYN**
 - d. **03_GATE**
 - e. **09_SUBMIT**

Design Description



The Transformer model, was the first to propose the self-attention mechanism as its core component.

Self-attention allows the model to capture long-range dependencies more effectively than previous architectures like RNNs and CNNs.

This innovation led to significant advancements in NLP, forming the foundation for models like GPT.

The Picture is from paper:

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017).

Attention is all you need.

arXiv url: <https://arxiv.org/abs/1706.03762>

For this design, we're going to design an simplified Two-Head Attention module, with five tokens and four dim of each word embedding in Verilog code.

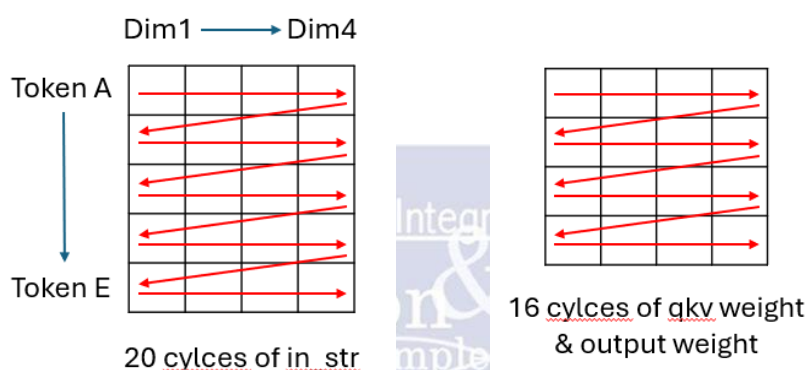
Ps. for real gpt:

- ✧ word embedding dim can range from 768 for GPT2 to more than 10^3 for GPT4
- ✧ head num can range from 12 for small GPT2 to more than 128 for GPT4

In the following contents, we will introduce the architecture of our attention module step by step, but for more information about the principles of self-attention, you can search online, there are many resources and courses about it.

The in_valid is pulled high for 20 cycles, and you will get

1. 20 cycles of in_str in raster scan order (5 tokens x 4 dimensions)
2. 16 cycles of k, q, v's weight, and the out_weight for output linear layer (4 x 4)

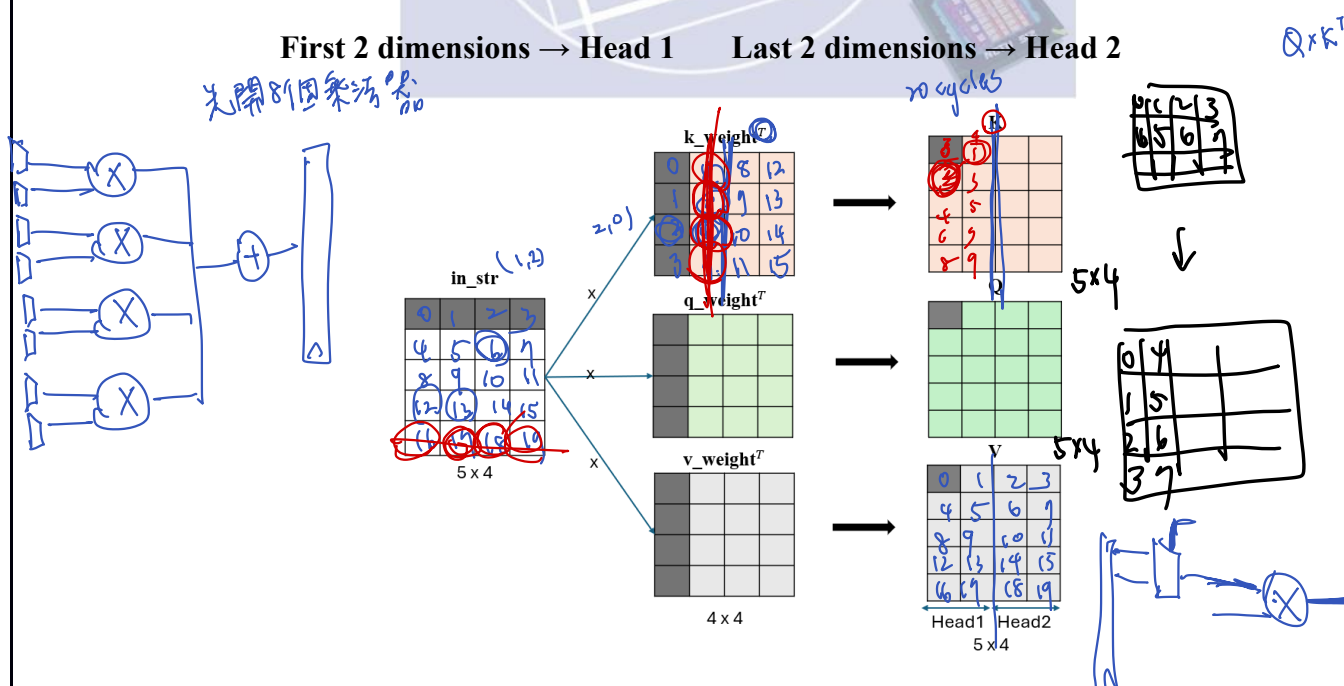


In self-attention, each token is associated with three weight matrices: **Key (K)**, **Query (Q)**, and **Value (V)**. The computation follows these steps:

1. Compute K, Q, V: Multiply the input embedding matrix with the respective weight matrices:

$$K = \text{in_str} \times k_weight^T \quad Q = \text{in_str} \times q_weight^T \quad V = \text{in_str} \times v_weight^T$$

2. You will have 4 dims for each K, Q, V. Since we use **2 heads**, we split them:



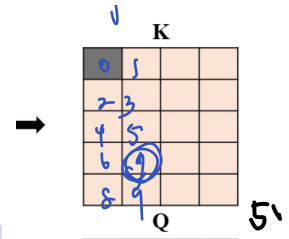
3. For each head, do the following computations:

a. Score : compute the attention scores by $Q \times K^T$

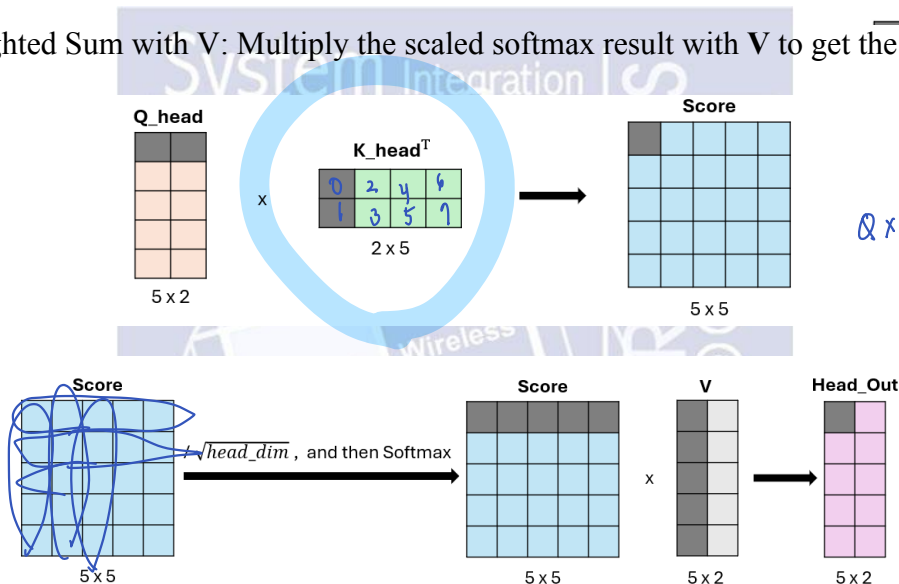
b. Head Scaling: to make sure the softmax_res in a reasonable range, each number will be divided by $\sqrt{\text{head_dim}}$, which is $\sqrt{2}$

`parameter square_root_2 = 32'b00111111101101010000010011110011;`

c. Softmax(Score) : do the softmax for each row of Score

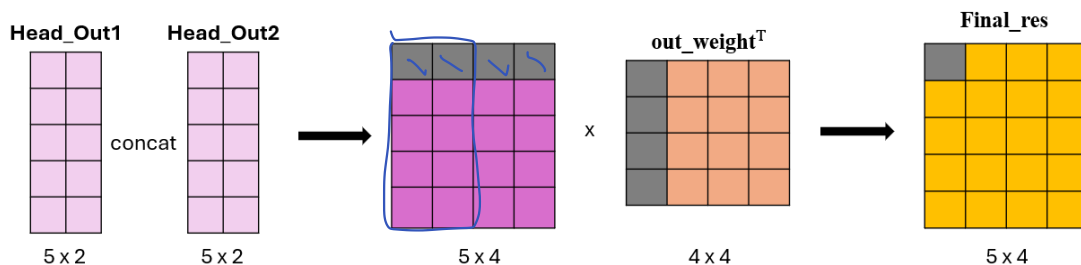


d. Weighted Sum with V: Multiply the scaled softmax result with V to get the head output



4. Concatenation: concat the head1 to the first two dim, and head2 for the last 2 dim

5. Linear Projection: Pass through a **linear layer** to project back to the original embedding space.



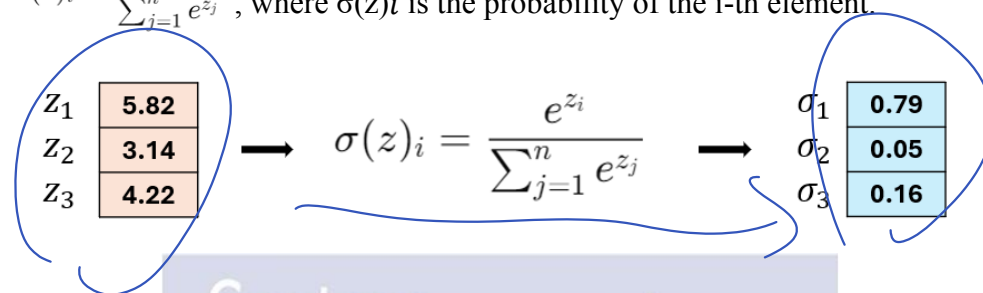
6. Output your Linear Projection results in raster scan order, so you have to output for 20 cycles.

■ Softmax:

Softmax applies an exponential function to each score and normalizes them, representing the probability of a class being the correct one.

For a vector $z = [z_1, z_2, \dots, z_{n-1}, z_n]$, the Softmax function for the i -th element is defined as:

$$\sigma(z)_i = \frac{e^{z_i}}{\sum_{j=1}^n e^{z_j}}, \text{ where } \sigma(z)_i \text{ is the probability of the } i\text{-th element.}$$



Inputs and Outputs

The following are the definitions of input signals

Input Signals	Bit Width	Definition
clk	1	Clock.
rst_n	1	Asynchronous active-low reset.
in_valid	1	High when all input is valid.
in_str	32	The token signals which sent in raster order. The arithmetic representation follows the IEEE-754 floating number format. (Range: $\mp 0 \sim 0.5$)
k_weight	32	The weight signals which sent in raster order. The arithmetic representation follows the IEEE-754 floating number format. (Range: $\mp 0 \sim 0.5$)
q_weight	32	The weight signals which sent in raster order. The arithmetic representation follows the IEEE-754 floating number format. (Range: $\mp 0 \sim 0.5$)
v_weight	32	The weight signals which sent in raster order. The arithmetic representation follows the IEEE-754 floating number format. (Range: $\mp 0 \sim 0.5$)

out_weight	32	The weight signals which sent in raster order. The arithmetic representation follows the IEEE-754 floating number format. (Range: $\mp 0\sim 0.5$)
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The following are the definitions of output signals

Output Signals	Bit Width	Definition
out_valid	1	High when out is valid.
out	32	The output signals which will be sent in raster scan order for the final output. The arithmetic representation follows the IEEE-754 floating number format.

Each time you output the result, our pattern will check the correctness of it. Basically, if you follow the formulas and use IEEE floating point number IP, you should get same result as our answer.

However, we release the constraint; you may have an error under 10e-7 for the result after converting to float number. This means that we will convert your output from binary format into real float number, and compare with our answer. Error will be calculated by '(golden-ans)/golden' and get its absolute value. If the error is higher than the value, you will fail this lab.

Ex:

Binary form: 00111101101100101010000001000101

IEEE floating number: 0.08721975

nWave:

8.72198e-02

nWave will round the number for display, but the computation will not be affected. The following numbers are all from nWave, thus the computations are performed by IPs in Verilog.

※ $e-02 = 10^{-2}$

- [illegible]

$$\text{value} = (+1) \times 1.25 \times 2^{-3} = +0.15625$$

- ## Specifications

- 6 -

10. The output delay is set to **0.5*(clock period)**, and the output loading is set to **0.05**.
11. The synthesis result of data type **cannot** include any **latches**.
12. After synthesis, you can check ATTN.area and ATTN.timing. The area report is valid when the slack in the end of timing report should be **non-negative (MET)**.
13. **In this lab, you must use at least one IEEE floating point number IP from Designware. We will check it at ATTN.resource in 02_SYN/Report/. The example shows in following figure.**

```

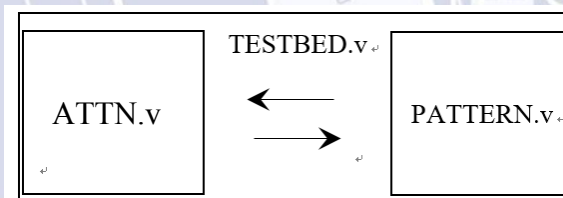
=====
| Cell      | Module      | Parameters | Contained Operations |
=====
| add_x_5   | DW01_inc    | width=4    | add_194 (CNN.v:194)  |
| add_x_6   | DW01_inc    | width=11   | add_207 (CNN.v:207)  |
| lt_x_7    | DW_cmp      | width=11   | lt_230 (CNN.v:230)   |
| lt_x_8    | DW_cmp      | width=11   | lt_237 (CNN.v:237)   |
| M0        | DW_fp_mult  | sig_width=23 | M0 (CNN.v:1741)      |
|           |             | exp_width=8 |                       |
|           |             | ieee_compliance=0 |                       |
| M1        | DW_fp_mult  | sig_width=23 | M1 (CNN.v:1742)      |
|           |             | exp_width=8 |                       |
|           |             | ieee_compliance=0 |                       |

```

Grading Policy

1. Function Validity: 70%
2. Performance: 30 %
 - Area * Computation time: 30%
 - Computation time = Latency * clock cycle time

Block diagram



Note

1. Please submit following files under 09_SUBMIT before 12:00 at noon on Mar. 24:
 - ATTN.v
 - If uploaded files **violate the naming rule**, you will get **5 deduct points**.
 - In this lab, you can adjust your clock cycle time. **Consequently, make sure to key in your clock cycle time when you submit**. It's means that the TA will demo your design under this clock cycle time.
 - The 2nd demo deadline is **12:00 at noon on Mar. 26**.
 - Check whether there is any wire / reg / submodule name called "error", "fail", "pass", "congratulation", "latch", "DW_fp", if you used, you will fail the lab.

2. Template folders and reference commands:

01_RTL/ (RTL simulation) `./01_run_vcs_rtl`

02_SYN/ (Synthesis) `./01_run_dc_shell`

(Check if there is any **latch** in your design in **syn.log**)

(Check the timing of design in **/Report/ATTN.timing**)

03_GATE / (Gate-level simulation) `./01_run_vcs_gate`

✳️ You should make sure the three clock period values identical in 00_TESTBED/PATTERN

`.v && /02_SYN/syn.tcl:`

```
`define CYCLE_TIME      50.0
`define SEED_NUMBER     28825252
`define PATTERN_NUMBER 1000
```

```
#=====
# (A) Global Parameters
#=====
set DESIGN "ATTN"
set CYCLE 50
```

Sample Waveform

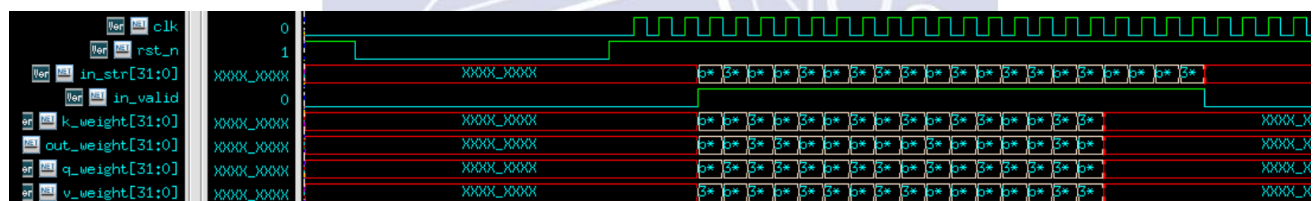


Fig1. Input waveform

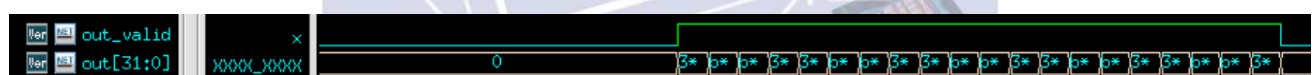


Fig2. Output waveform